Rony Zeenaldeen

Backend & Distributed Systems Engineer

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SUMMARY

Backend & Distributed Systems Engineer with 4+ years of experience designing and building production-grade backend systems in Go (Golang), Python and Java (Spring Boot). Experienced in event-driven architectures, microservices, multi-tenant platforms and live AI integration pipelines, backed by PostgreSQL, MySQL, Redis, Kafka and RabbitMQ, with infrastructure-as-code deployment and Prometheus/Grafana observability. Proven impact: PostgreSQL query times cut from ~2000ms to under 50ms (40× faster), 60–70% API latency reduction via Redis caching, and 65% lower database load.

TECHNICAL SKILLS

Languages: Go (Golang), Python, Java, JavaScript, SQL

Back-End: Django/DRF, FastAPI, Spring Boot, Node.js, NestJS, Microservices, REST APIs, GraphQL, WebSockets, OpenAPI

Distributed Systems & Messaging: Apache Kafka, RabbitMQ, Celery, Event-Driven Architecture, Asynchronous Workflows

Databases: PostgreSQL, MySQL, Redis, MongoDB, pgvector

DevOps & Infrastructure: Docker, Kubernetes, Helm, k3s, Terraform, Ansible, NGINX, GitHub Actions, CI/CD, Linux

Cloud: AWS (IAM, S3, RDS, SES, Route 53, Elastic Beanstalk, EC2, ECS Fargate)

Observability & Reliability: Prometheus, Grafana, pgBackRest, pgBadger

Additional: Flutter, Next.js, React

PROFESSIONAL EXPERIENCE

AI & Backend Engineer (Independent)

Nov 2024 – Present

- **Enterprise Document Automation Backend (Client – Austria / Remote):** Engineered an end-to-end Django pipeline converting scanned Arabic documents to formatted DOCX, collapsing a multi-stage OCR/LLM stack into one strict-JSON VLM call to eliminate error compounding — overriding python-docx limits via direct OpenXML table-cell (w:tc) manipulation and run-merging to preserve exact fonts and layouts.
- **Event-Driven Infrastructure & Service Architecture:** Provisioned multi-node IaC environments (Terraform, Ansible) running decoupled FastAPI microservices across Kafka/RabbitMQ pipelines, with Prometheus/Grafana observability and pgBackRest point-in-time recovery.
- **Workflow Routing & Role-Based Access Control:** Built a confidence-scored (0.0–1.0) routing workflow that auto-passes clean fields and routes uncertain ones (<0.85) to role-based reviewers (Employee/Reviewer/Admin), with full audit logging.
- **Model-Serving APIs:** Shipped a chest X-ray classifier (EfficientNetV2S, 91–92% accuracy / F1 0.95) behind a Django REST API, a Facebook Prophet sales-forecasting pipeline on 1M+ transactions, and an HR attrition model (87% accuracy, $225K–$450K est. annual retention savings).

Backend Systems Engineer (Independent)

Jan 2022 – Nov 2024

- **Backend Performance Consulting — Qader (qader.vip):** Delivered independent Django performance tuning for a production Saudi learning-management platform, profiling slow endpoints and optimizing ORM queries and database access to improve response times under load.
- **Performance & Caching:** Optimized PostgreSQL query response times from 2000ms to <50ms, cut database load by 65%, and reduced API latency by 60–70% through strategic Redis caching.
- **API Design & Integrations:** Built and shipped secure REST APIs (Django REST Framework, Go) with JWT/OAuth2 auth, versioning, and OpenAPI docs, plus GraphQL endpoints and Stripe/PayPal/Razorpay payment integrations with secure webhook handling.
- **Real-Time & Distributed Systems:** Designed real-time systems (WebSockets, Django Channels, Redis) for live ride-matching and location streaming; deployed Go microservices for latency-critical, high-concurrency components, with RabbitMQ/Kafka decoupling and pgBackRest/pgBadger strengthening PostgreSQL backup and query analysis.
- **Cloud & DevOps:** Deployed backend services using Docker, Kubernetes, NGINX, and AWS infrastructure for high availability and scalability.

KEY PROJECTS

Mizan AI — Enterprise Multi-Tenant SaaS & ERP Platform (LIVE) · 2026 ↗ Link

Production-grade financial & inventory platform for retail operations with live AI pipelines and real-time observability.

- **Modular Monolith:** Engineered a high-performance Java 21 Spring Boot 3 modular monolith backend paired with a Next.js 16 frontend and PostgreSQL/Redis storage layers.
- **Live AI Integration:** Built production pipelines for OpenAI whisper-1 (voice transaction capture) and gpt-4o-mini (receipt OCR and auto-bookkeeping) with strict currency/decimal precision.
- **DevOps & Observability:** Deployed on a Linux VPS via Docker & an Nginx reverse proxy; integrated a Prometheus & Grafana monitoring stack for real-time latency, memory and HTTP metrics.

Tech: Java 21 · Spring Boot 3 · Next.js 16 · PostgreSQL · Redis · Docker · Nginx · Prometheus · Grafana · OpenAI API

Multi-Tenant & Multi-Domain Enterprise CMS (LIVE)

Live: ↗ zenex.nl · ↗ global-transport.nl · ↗ as-zen-logistiek.nl · ↗ rit-bv.nl

Scalable CMS platform powering 4 independent Dutch enterprise production domains from a single Django backend.

- **Multi-Tenant Architecture:** Engineered a multi-tenant, multi-domain Django backend serving 4 independent Dutch enterprise domains with strict tenant data isolation, dynamic per-domain routing and scalable request handling.
- **Dynamic Dashboards & RBAC:** Built fully dynamic management dashboards with granular Role-Based Access Control (RBAC) over content and administrative actions.
- **Data Layer & Deployment:** Optimized PostgreSQL querying and Redis caching for multi-domain state management, served through Dockerized services behind an NGINX reverse proxy.

Tech: Python · Django · PostgreSQL · Redis · Docker · NGINX

NeuroSeek AI — Agentic AI Research Platform (LIVE) · 2026 – Present ↗ Link

Self-hosted research platform on a polyglot, event-driven backend, streaming fully cited RAG reports token-by-token.

- **Polyglot Service Architecture:** Engineered a Django 5 + DRF + Celery/Redis backend, a Go WebSocket streaming hub and a Kafka (KRaft) consumer, streaming cited reports live to web and a native cross-platform mobile client.
- **Infrastructure & Reliability:** Terraform/Ansible on Docker Compose/Kubernetes (Helm, k3s), Prometheus/Grafana and pgBackRest (PITR)/pgBadger; hardened with SSRF protection, JWT rotation/blacklisting and rate limiting.
- **Agentic Retrieval:** A “gap-driven” loop audits answer coverage and re-searches before finalizing a report.

Tech: Go · Python · Django · DRF · Celery · Redis · PostgreSQL · pgvector · Kafka · Ollama · Terraform · Ansible · Kubernetes · Prometheus

PulmoSight — Doctor-in-the-Loop Chest X-Ray AI (LIVE) · 2026 – Present ↗ Web App · ↗ Mobile App

Self-hostable clinical system for 18 chest pathologies, with separated application, inference and delivery tiers.

- **Service Separation:** Django 5 + DRF application plane, an independent FastAPI/PyTorch inference service and a Go Telegram bot, decoupled over RabbitMQ so inference and delivery restart independently of the web workflow.
- **Secure Delivery:** E-signed PDFs move through expiring, dual-salted capability tokens via a PIN-verified Go bot.
- **Mobile Client Integration:** Integrated a role-aware native Flutter client (v1.1.0) against the same API surface, with hardware-backed keystore tokens, AES-encrypted offline storage and biometric locks.

Tech: Python · Django 5 · DRF · FastAPI · Go · RabbitMQ · PostgreSQL · PyTorch · Docker · Flutter · Telegram Bot API

OPEN SOURCE CONTRIBUTIONS

- **Official Contributor to pydantic-settings (PR #920):** Fixed generic-subclass detection breaking Secret-type env-var loading, merged and shipped in v2.15.0; Issue #10177 · Contributor Graph.
- **FastAPI (PR #12066):** Reproduced a concurrent-load dependency-teardown deadlock (AnyIO threadpool exhaustion).
- **langchaingo (PR #1513):** Fixed the Anthropic handler dropping tool-call content on multi-part AI messages.

EDUCATION

B.Sc. Informatics Engineering — Software & Applications Development Sep 2020 – Jun 2025

Al-Shahba Private University, Aleppo, Syria

CERTIFICATIONS

- Artificial Intelligence Essentials V2 — IBM (2025) [Link]
- Data Engineering Professional — IBM (2025) [Link]
- Strategic Planning Professional Certificate — Canada [Link]
- Project Management Professional (PMP v6) — Project International (2021) [Link]

LANGUAGES

- Arabic — Native
- English — B2-C1 (Proficient)